

FLIPKART E-COMMERCE DATABASE MANAGEMENT SYSTEM

Skill Task: DBMS

1.Database Design

1.1 Category Table

Entity: category

Attribute	Key type
category_ID	Primary key
Category_Name	Unique key
Description	-

1.2 Product Table

Entity: product

Attribute	Key type
Product_ID	Primary key
Product_name	Not Null
Category_ID	Foreign key
Seller_ID	-
brand	-
price	-
stock	-
Description	-

2. PRIMARY AND FOREIGN KEY RELATIONSHIP

Category → Product

- **Category.Category_ID → Primary Key**
- **Product.Product_ID → Primary Key**
- **Product.Category_ID → Foreign Key referencing Category.Category_ID**
- **Relationship: One-to-Many (1:M)**

3. Table Creation

3.1 Create Category Table

```
SQL> CREATE TABLE Category (  
2     Category_ID NUMBER PRIMARY KEY,  
3     Category_Name VARCHAR2(50) UNIQUE NOT NULL,  
4     Description VARCHAR2(100)  
5 );
```

Table created.

3.2 Create Product Table

```
SQL> CREATE TABLE Product (  
2     Product_ID NUMBER PRIMARY KEY,  
3     Product_Name VARCHAR2(100) NOT NULL,  
4     Category_ID NUMBER,  
5     Price NUMBER(10,2),  
6     Stock NUMBER,  
7     Seller_ID NUMBER,  
8     FOREIGN KEY (Category_ID) REFERENCES Category(Category_ID)  
9 );
```

Table created.

4. Data Insertion

4.1 Category Records

```
SQL> INSERT INTO Category VALUES  
2 (1, 'Electronics', 'Electronic products');  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO Category VALUES  
2 (2, 'Fashion', 'Clothing and fashion products');  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO Category VALUES  
2 (3, 'Home Appliances', 'Appliances for home');  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO Category VALUES  
2 (4, 'Books', 'Books and educational products');  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO Category VALUES  
2 (5, 'Beauty', 'Beauty and personal care products');  
  
1 row created.
```

4.2 Product Records

```
SQL> INSERT INTO Product VALUES
  2 (101, 'Laptop', 1, 55000, 20, 201);

1 row created.

SQL>
SQL> INSERT INTO Product VALUES
  2 (102, 'Smartphone', 1, 25000, 30, 202);

1 row created.

SQL>
SQL> INSERT INTO Product VALUES
  2 (103, 'T-Shirt', 2, 799, 50, 203);

1 row created.

SQL>
SQL> INSERT INTO Product VALUES
  2 (104, 'Mixer Grinder', 3, 2500, 15, 201);

1 row created.

SQL>
SQL> INSERT INTO Product VALUES
  2 (105, 'Python Programming Book', 4, 650, 25, 204);

1 row created.

SQL>
SQL> INSERT INTO Product VALUES
  2 (106, 'Face Wash', 5, 350, 40, 205);

1 row created.
```

5. Data Retrieval

5.1 Display Category Records

```
SQL> Select * From Category;
```

CATEGORY_ID	CATEGORY_NAME	DESCRIPTION
1	Electronics	Electronic products
2	Fashion	Clothing and fashion products
3	Home Appliances	Appliances for home
4	Books	Books and educational products
5	Beauty	Beauty and personal care products

5.2 Display Product Records

```
SQL> Select * From Product;
```

PRODUCT_ID	PRODUCT_NAME	CATEGORY_ID	PRICE	STOCK	SELLER_ID
101	Laptop	1	55000	20	201
102	Smartphone	1	25000	30	202
103	T-Shirt	2	799	50	203
104	Mixer Grinder	3	2500	15	201
105	Python Programming Book	4	650	25	204
106	Face Wash	5	350	40	205

6 rows selected.

6. Update Product

update the price of the Laptop:

```
SQL> UPDATE Product
  2 SET Price = 52000
  3 WHERE Product_ID = 101;

1 row updated.
```

7. Product Deletion

Delete the Product face Wash

```
SQL> DELETE FROM Product
  2 WHERE Product_ID = 106;

1 row deleted.
```

8. CATEGORY-WISE PRODUCT REPORT

8.1 Display products with their category names:

```
SQL> SELECT
  2 c.Category_Name,
  3 p.Product_Name,
  4 p.Price,
  5 p.Stock
  6 FROM Category c
  7 JOIN Product p
  8 ON c.Category_ID = p.Category_ID
  9 ORDER BY c.Category_Name;
```

CATEGORY_NAME	PRODUCT_NAME	PRICE	STOCK
Books	Python Programming Book	650	25
Electronics	Laptop	52000	20
Electronics	Smartphone	25000	30
Fashion	T-Shirt	799	50
Home Appliances	Mixer Grinder	2500	15

8.2 PRODUCTS IN ELECTRONICS CATEGORY

```
SQL> SELECT
  2     p.Product_ID,
  3     p.Product_Name,
  4     p.Price,
  5     p.Stock
  6 FROM Product p
  7 JOIN Category c
  8 ON p.Category_ID = c.Category_ID
  9 WHERE c.Category_Name = 'Electronics';
```

PRODUCT_ID

PRODUCT_NAME

PRICE STOCK

101
Laptop 52000 20

102
Smartphone 25000 30

PRODUCT_ID

PRODUCT_NAME

PRICE STOCK

8.3 PRODUCT COUNT BY CATEGORY

```
SQL> SELECT
  2     c.Category_Name,
  3     COUNT(p.Product_ID) AS Total_Products
  4 FROM Category c
  5 LEFT JOIN Product p
  6 ON c.Category_ID = p.Category_ID
  7 GROUP BY c.Category_Name
  8 ORDER BY c.Category_Name;
```

CATEGORY_NAME TOTAL_PRODUCTS

Beauty 0
Books 1
Electronics 2
Fashion 1
Home Appliances 1